Detector production pilot: formulas and results

NAUSICAA generated report

This calculation uses the recorded parameter cards. The approximate J values at $E/P = 1.5, 2, 3$ are user readings of Oliveira et al., Figure 6; the form requires an explicit J at intermediate fields.

Resolved inputs

Parameter	Value
TPC length	150.0 cm
Pressure	15.0 bar
Ion drift field	200.0 V/cm
Ion pitch, frame	3.0 mm, 30.0 ms
Electron hole pitch, bucket	10.0 mm, 2.0 μs
Electron lifetime	15.0 ms
EL E/P	3.00 kV/cm/bar
Pair-creation energy W, Fano factor F	21.90 eV, 0.150
EL variance ratio J	0.010
J source	oliveira_fig6_user_reading

Calculations and assumptions

$$\mu_i = 10^6 E_i/W, \qquad \mathrm{Var}(N_i) \simeq F\mu_i,$$
$$L_e = L_{\max} - L_i, \qquad L_i \sim U(L_{\min}, L_{\max}),$$
$$\sigma_{i,T} = \sigma_{i,L} = 10\sqrt{2(k_B/e)TL_i/E_d}\ \mathrm{mm},$$
$$\sigma_{e,T/L} = d_{T/L}\sqrt{L_e/P}, \qquad p_{\mathrm{survive}} = e^{-L_e/(v_e\tau_e)},$$
$$Y_g = a_g \max(E/P - x_{0,g}, 0)Ph, \qquad \mathrm{Var}(N_{\mathrm{EL}}|n_e) = n_e J Y_g,$$
$$A = (h + r - \sqrt{h^2 + r^2})/(2h), \qquad N_{\mathrm{incident}}|N_{\mathrm{EL}} \sim \mathrm{Binomial}(N_{\mathrm{EL}}, A).$$

At each source hit, one integer pair draw creates both initial carrier populations. The minimum integer variance limits the Fano target at low occupancy. Electron attachment is sampled once per pair realisation and shared by the two gas alternatives. Diffused carriers enter physical grid cells directly; there is no diffusion-window charge loss. VUV photons touching a hole wall are absorbed; a circular SiPM covers its exit.

Derived quantity	Pure Xe	Xe–15% He
Max transverse electron σ (mm)	11.068	5.060
Max longitudinal electron σ (mm)	2.846	2.372
Mean EL yield (photons/electron)	2,332.4	2,293.5

Ion width at maximum drift: 1.947 mm. Ion frame: 3.0 mm. Electron bucket: 2.0 mm. Minimum electron survival: 0.9048. Direct optical acceptance: 0.19098. Gate/anode potentials relative to cathode: 30.0 / 52.5 kV.

Saved event totals and validation

Sample	Events	Pairs	Ions	Surviving e^-	Xe photons	Xe–He photons
bb0nu	8,000	897,306,359	897,306,359	856,854,917	381,687,642,280	375,319,166,815
single_electrons	8,000	897,244,198	897,244,198	856,968,162	381,739,047,046	375,368,751,796

The ion ledger is exact for both samples: ions in pad/time cells equal initial pairs. For each gas, incident photons do not exceed emitted photons. The three branch files store matching per-event pair checksums.

Provenance and scope

Run ID: `238ddc5a75e0fd9b`. Code SHA-256 prefix: `803fbbe616ed98ec`. The complete digest is in the HDF5 summary. Peak process RSS: 1069.2 MB. Source hit tables are the 1 mm Geant4 samples. Ion thermal diffusion follows ITACA Revisited, Eq. (2.1) (arXiv:2604.02443). The pure-Xe transverse coefficient is its Table 1 input; other diffusion coefficients are recorded model inputs. EL yield coefficients follow Fernandes et al., JHEP 04 (2020) 034, Table 1. The definition of J follows Oliveira et al., JINST 6 (2011) P05007, Eq. (4.6); the recorded field-point values are approximate user readings of its Figure 6.